UGANDA ADVANCED CERTIFICATE OF EDUCATION
APPLIED MATHEMATICS PAPER 2
COMPETENCY-BASED ASSESSMENT ITEMS

Time: 3 Hours

INSTRUCTIONS TO CANDIDATES:

- This paper consists of three sections: **A** (Mathematical Statistics), **B** (Mechanics), and **C** (Numerical Methods).
- Answer **five** items in total: at least **one** item from Section A, at least **one** item from Section B, and at least **one** item from Section C.
- All items carry equal marks. Show all mathematical modeling, structural equations, flowcharts, and final interpretations clearly.
- In numerical calculations, take the acceleration due to gravity, g, to be 9.8 m s^{-2}.

SECTION A: MATHEMATICAL STATISTICS

*Answer at least **one** item from this section.*

ITEM 1 (Agricultural Quality Control & Crop Yield Modeling)

Scenario:
A large agricultural cooperative in Jinja conducts quality control audits on seed distribution bags before supply to regional smallholder farmers. Initial processing facilities sort mixed production batches containing standard seed grains, high-yield organic hybrid grains, and rare drought-resistant premium variations. Inspectors need to determine sampling configurations from small aggregate seed selections to minimize regional distribution errors (Montgomery, 2019). Simultaneously, the cooperative tracks long-term regional harvest yields across a large sample of municipal farming zones. Agronomists observe that total crop biomass weights under optimal weather seasons follow a highly stable continuous probability distribution model, allowing supervisors to predict the likelihood of meeting minimum export thresholds and avoid supply shortages (Al-Kaisi & Kwaw-Mensah, 2020).

Tasks:

a) If a small validation batch contains 6 standard seed grains, 3 organic hybrid grains, and 1 premium grain, calculate the exact mathematical probability that a random selection of three grains gathered without replacement contains exactly 2 standard seed grains and 1 organic hybrid grain.

b) The operational yield function across a major municipal zone is governed by a continuous random variable X spanning between boundary parameters 2 and 5. The cumulative distribution function for $X \leq x$ is modeled by $ax - \frac{b}{x^2}$. Determine the exact design constants a and b, extract the probability density function (pdf) of X, and compute the exact mean regional crop yield.

c) Calculate the explicit tracking probability $P(3 < X < 4)$ to evaluate target mid-tier agricultural production stability thresholds.

ITEM 2 (Academic Performance Analytics & District Educational Forecasting)

Scenario:
A district education task force evaluates performance metrics across rural academic centers to coordinate resource allocations and teacher development grants. Analysts gather aggregate final grades from a district-wide mock examination to assess overall student proficiency levels, establishing benchmarks to measure the probability of distinction passes or failures across regional secondary institutions (Sutherland et al., 2021). Concurrently, the advisory board evaluates the core relationship between internal language arts proficiency scores and conceptual historical literacy evaluations among students. By analyzing student performance variations across these two distinct subject matrices, administrators look to quantify curriculum symmetry and determine if secondary training formats show strong positive skill transfers (Goldstein, 2021).

Tasks:

a) The final grade scores gathered from 1000 district candidates follow a stable normal distribution model with a mean score of 55 and a standard deviation of 8. If an advanced merit certificate requires scoring 71 marks or more, estimate the total number of students awarded certificates.

b) Given that exactly 15% of the total district candidates failed the final evaluation, determine the minimum passing score threshold. Find the probability that two randomly selected candidates both passed the examination.

c) The comparative language arts (x) and historical literacy (y) marks for a core evaluation panel are tabulated below:

Language (x):	13	15	15	29	20	20	21	21	24
History (y):	65	60	76	62	70	75	76	80	70

Calculate the exact Spearman rank correlation coefficient for this metric panel, and provide a detailed academic commentary regarding the performance relationship between these subjects.

ITEM 3 (Athletic Training Logistics & Health Metrics Profiling)

Scenario:
A regional sports science facility tracks the weekly training workloads of long-distance

runners to optimize cardiovascular conditioning profiles and prevent overtraining syndrome. Medical trainers compile the total distances covered by club athletes over an initial preparatory cycle, creating structured frequency matrices to analyze internal demographic spreads and check compliance with target fitness benchmarks (Cobb, 2020). To evaluate target milestone compliance, the coaching staff needs to construct spatial tracking curves that model cumulative training metrics. These visualization curves allow analysts to locate median thresholds, check variability across distinct performance quartiles, and track the volume of elite runners meeting extended endurance goals (Boring, 2019).

Tasks:

a) The weekly distance tracking parameters recorded across the athletic cohort are summarized in the table below:

Distance (km):	31–40	41–45	46–50	51–55	56–57	58–60	61–70	71–90
Frequency:	10	15	20	70	64	24	20	10

Calculate a precise statistical estimate of the standard deviation of training distances to measure performance consistency across the club.

b) Using the provided distribution table, construct an accurate cumulative frequency curve (Ogive) on standard graphical coordinates.

c) From your completed Ogive curve, estimate the semi-interquartile range of the cohort workload, and determine the exact number of regional athletes who completed training distances between 50.0 km and 66.0 km.

SECTION B: MECHANICS

*Answer at least **one** item from this section.*

ITEM 4 (Automotive Powertrain Dynamics & Structural Friction Mechanics)

Scenario:
A mechanical engineer evaluates the heavy-duty powertrain configuration of an electric transport vehicle carrying cargo along inclined delivery routes. The cargo unit must accelerate smoothly from rest up an inclined operational plane without slipping backwards, requiring the vehicle's motor to match resistive forces and weight components to reach maximum speed (Genta & Morello, 2020). In another division of the maintenance laboratory, technicians measure structural safety parameters for static machinery blocks anchored along loading docks. Mechanical components are exposed to directional tension cables pulling at steep angles. Safety teams need to verify the operational friction coefficients between material surfaces to prevent slipping accidents and ensure structural anchorage zones remain stable (Hibbeler, 2022).

Tasks:

a) A transport vehicle of mass 1000 kg accelerates uniformly from rest up a smooth plane inclined at an angle of 30° to the horizontal until it reaches a maximum velocity of 90 km h^{-1}. Compute the maximum mechanical power output developed by the vehicle engine during this stabilization sequence.

b) A structural component of mass 10 kg rests on a rough horizontal surface and is subjected to an explicit directional cable tension force of magnitude $\frac{98}{\sqrt{3}}$ N directed at an angle of 60° above the horizontal. Given that the component remains perfectly stationary, calculate the minimum allowable value for the static coefficient of friction (μ) between the block and the support surface.

ITEM 5 (Industrial Cargo Logistics & Avionics Vector Navigation)

Scenario:
An industrial automated warehousing system uses overhead pulley tracks to move heavy freight bundles across primary assembly stations. Cable lines are connected to distinct load masses hanging vertically from support frames. When these sorting lines pick up or release component masses during transit, engineers must analyze structural shock loads and sudden changes in system acceleration to ensure smooth automated handling (Spong et al., 2020). Concurrently, a cargo flight coordinator models cross-country navigation lines for commercial logistics drones flying between regional hubs. The drone flight routes cross high-altitude wind currents that push the aircraft off course, requiring flight navigation systems to continuously adjust steering vectors to ensure precise arrival times (Anderson, 2021).

Tasks:

a) Two freight masses of 4 kg and 3 kg are attached to the opposite ends of a light, flexible, inextensible cable that passes over a small, frictionless overhead pulley. Assuming both hanging string sections remain vertical, determine the initial acceleration of the moving system and calculate the continuous operational tension force in the cable.

b) While the 3 kg mass is moving vertically upwards at a velocity of 9 m s^{-1}, it picks up a stationary secondary coupling mass of 2 kg at a fixed loading point A, instantly forming a combined cargo block Q of mass 5 kg. Calculate the new immediate velocity of the combined system, and find the maximum vertical height above point A to which the block Q will rise.

c) A regional distribution hub B is located 763 km away from a takeoff depot A along a true bearing vector of 080°. A logistics drone with a cruising speed of 400 km h^{-1} in still air flies directly from A to B through a constant environmental wind blowing directly from the North at a speed of 70.5 km h^{-1}. Determine the exact compass heading direction the navigation system must steer to maintain a direct track.

ITEM 6 (Ballistic Trajectory Analytics & Structural Static Equilibrium)

Scenario:
A robotic ballistics system tests safety netting designs by launching test projectiles toward defensive structural boundaries. Projectiles are fired from elevated platforms near retaining walls, requiring engineers to calculate exact parabolic equations of motion to ensure test objects clear containment shields safely without damage (Meriam et al., 2020). Simultaneously, civil engineers design concrete retaining walls using isosceles trapezoidal geometries to distribute static pressure loads across foundational supports. Five primary structural load points act along the geometric boundaries of the frame, requiring structural teams to use vector resolution to calculate the net resultant load force and track its structural center of pressure along the foundation base (Beer et al., 2020).

Tasks:

a) A test projectile is launched from an initial coordinate point O located at a vertical height of 2 m above the ground floor, directed at an angle of 45° above the horizontal line. The projectile travels a horizontal distance of 4 m before clearing the top edge of a protective containment wall that stands 1 m high. Prove that the exact spatial trajectory pathway of the flying object is modeled by the parabolic equation $16y = 16x - 5x^2$.

b) Based on this trajectory framework, calculate the total horizontal distance from the containment wall to the exact impact point where the object hits the ground floor. Determine the final velocity vector magnitude and directional impact angle at the moment of landing.

c) An isosceles trapezoidal support frame $ABCD$ has boundary parameters $AD = DC = CB = 1$ m and a base length $AB = 2$ m. Five static load forces of magnitudes 1 N, 3 N, 5 N, 6 N, and $2\sqrt{3}$ N act along vector directions $\vec{AD}$, $\vec{DC}$, $\vec{CB}$, $\vec{BA}$, and $\vec{AC}$ respectively. Given that the net resultant force vector cuts through the extended baseline AB at an external anchorage intersection node X, calculate the complete magnitude and angular direction of this resultant force, and find the structural distance AX.

SECTION C: NUMERICAL METHODS

*Answer at least **one** item from this section.*

ITEM 7 (Biochemical Laboratory Calibration & Statistical Error Analysis)

Scenario:
A bio-chemical laboratory technician measures sample concentration profiles by tracking natural logarithm values across baseline calibration control metrics. Because physical testing limits restrict measurements to discrete data steps, technicians use mathematical interpolation routines to estimate intermediate values and reverse-engineer base values from exponential output profiles (Chapra & Canale, 2021). Concurrently, a quality control analyst evaluates error propagation values inside automated liquid-handling systems. The processing unit calculates fluid throughput fractions using rounded decimal coordinates, requiring the data team to evaluate extreme absolute and relative error boundaries to guarantee that cumulative processing variances remain within safe manufacturing tolerances (Burden et al., 2020).

Tasks:

a) The experimental control metrics recorded by the laboratory calibration instruments are tabulated below:

X:	5.0	5.2	5.4	5.7	6.0
$\ln X$:	1.609	1.647	1.686	1.740	1.792

Using linear interpolation or extrapolation routines, calculate the estimated value of $\ln(5.56)$, and compute the inferred value of $e^{1.575}$.

b) A digital system monitors a complex operational throughput constant governed by the mathematical expression:

$$\frac{0.38}{4.28} - \frac{0.30}{2.14}$$

Assuming every number in this expression has been rounded off precisely to its stated number of decimal places, calculate the absolute value of the expression. Determine the maximum possible absolute error bounds and find the maximum relative error percentage for this transaction.

ITEM 8 (Fluid Dynamic Intersections & Algorithmic Optimization Modeling)

Scenario:
A computational fluid dynamics engineer models wave intersection profiles inside a water treatment facility's blending channels. The overlapping wave pathways follow a non-linear relationship where geometric profiles cross sinusoidal variations, requiring the engineering team to use visual graphing techniques to locate initial root approximations (Mathews & Fink, 2019). To automate these calculations, a systems programmer creates an iterative

software algorithm based on Newton-Raphson optimization. The algorithm must automatically process user inputs, execute convergence loops within fixed error limits, and print the optimized output parameters. This routine is validated using dry-run verification matrices to confirm code stability before deployment (Knuth, 2021).

Tasks:

a) Sketch the concurrent geometric curves $y = x^2$ and $y = \sin 2x$ on the same coordinate axes for the spatial domain $0 \leq x \leq \frac{\pi}{2}$. From your visual intersection plot, extract an approximate real root for the equation $x^2 - \sin 2x = 0$ correct to one decimal place.

b) Using the Newton-Raphson optimization method alongside your initial root approximation from task (a), calculate the refined real root of $x^2 - \sin 2x = 0$ correct to two decimal places.

c) Prove that the formal Newton-Raphson iterative algorithm template designed to compute roots for the transcendental equation $e^{2x} + 4x - 5 = 0$ simplifies to the following expression:

$$x_{n+1} = \frac{e^{2x_n}(2x_n - 1) + 5}{2e^{2x_n} + 4}, \quad \text{for } n = 0, 1, 2, \ldots$$

d) Design a complete, professional programmatic flowchart that reads an initial numerical approximation x_0, iteratively processes the formula derived in task (c), terminates when the absolute error margin drops below 1.0×10^{-4}, and prints both the final root and the total iteration count.

e) Perform an explicit, step-by-step dry run evaluation of your flowchart using a starting value of $x_0 = 0.5$, showing all intermediate calculation columns clearly.